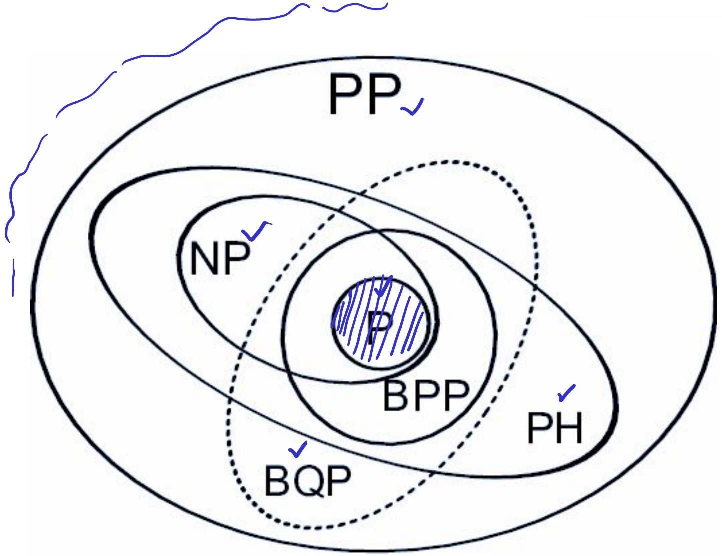
Complexity Classes: P, NP, NP-C, BQP

- **Polynomial Complexity Class** (Set)
  - Problems that can be solved in → $f(n) \in O(n^k)$
    → at-most polynomial time-steps in the size of input
  - Formally: **a complexity class is a set of languages**

$$P = \{L \subseteq \{0,1\}^* \,,$$
$$\text{there exists an algorithm } A_L \text{ that decides } L \text{ in polynomial time.}$$



Factoring

$N = (p \cdot q)$

- **Non-deterministic Polynomial Complexity Class**
  - Problems that can be verified in
    → at-most polynomial time-steps in the size of input

$$\mathrm{NP} = \{L \subseteq \{0,1\}^* \mid \text{ there exists a polynomial time verification algorithm } A \text{ that verifies } L\}$$

$O(n \cdot m)$

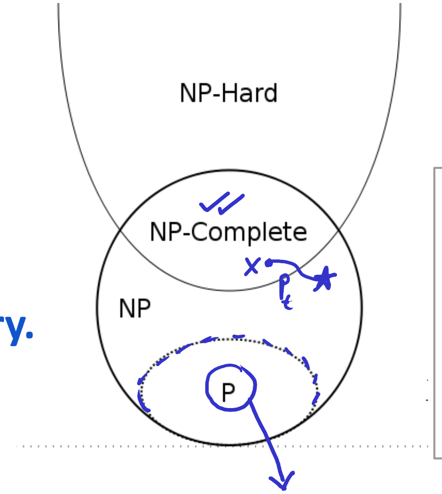
- **NP-Complete:** (X)

→ X is NP (can be verified in polynomial time-steps) (i)

→ Every problem in NP is polynomial-time reducible to X (ii)

→ **Reduction:** transform inputs and outputs such that the mapping is invariant for transformation

→ **NP-C problems are ubiquitous in science, technology, and industry.**



https://complexityzoo.net/Complexity_Zoo

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